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Disaster Alert & Community Response App

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Problem Statement

- During disasters, people may not receive timely alerts.
- Incident reporting and location sharing can be difficult.
- Affected people may struggle to find nearby assistance.
- Communication between citizens and response teams can be fragmented.
- A centralized platform is needed for faster information sharing and coordination.





Objectives

- Provide timely disaster alerts to users.
- Allow users to report emergency incidents with location.
- Enable users to request emergency assistance.
- Support volunteers and community members in helping affected people.
- Help users find nearby shelters and emergency services.
- Provide official announcements and updates during disasters.
- Enable map-based disaster tracking.
- Improve communication, coordination, and community response during emergencies.



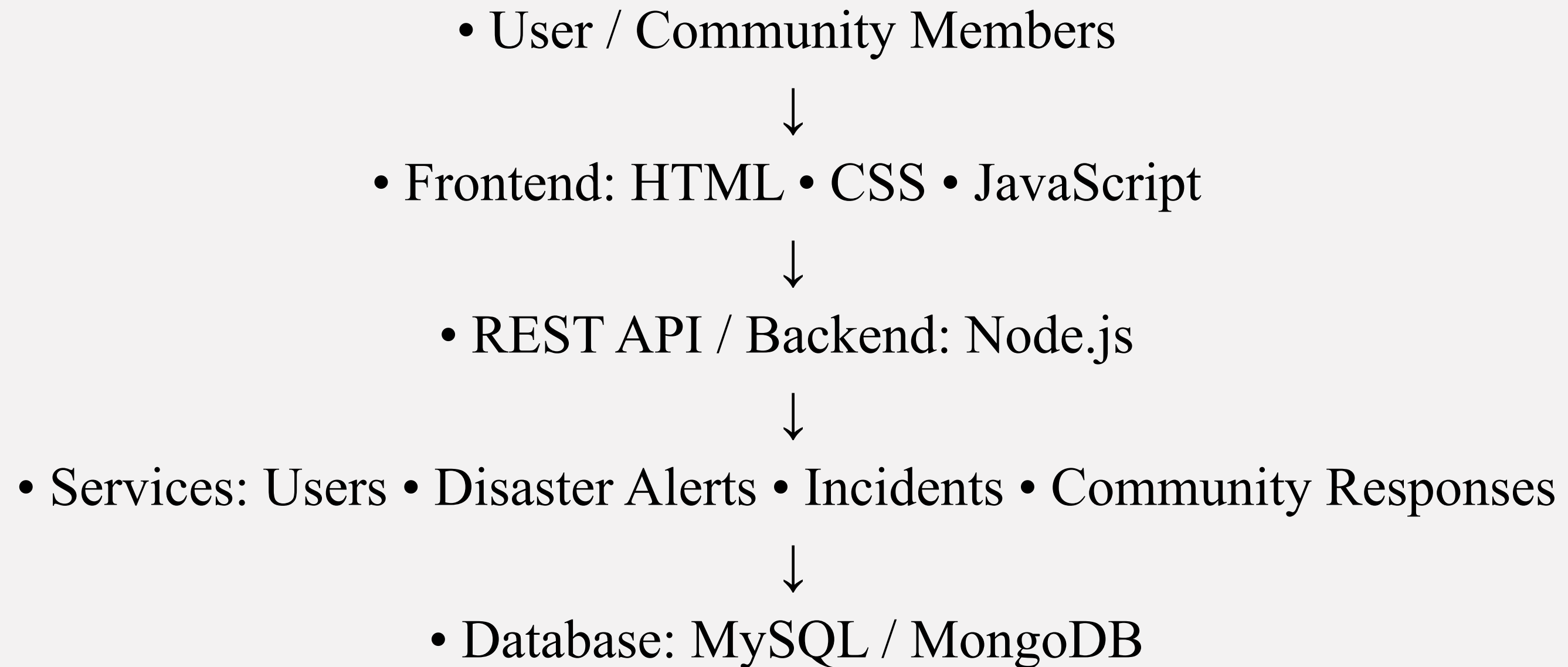


Proposed System

- Centralized disaster communication and response platform.
- Users can receive alerts, report incidents and request help.
- Community members can offer assistance to affected people.
- Map-based features support location-aware disaster tracking.
- Backend services manage users, alerts, incidents and responses.
- Authentication and authorization help protect user and emergency information.



System Architecture





Modules / Functionalities



1. User & Emergency Contact Management
2. Real-time Disaster Alerts
3. Incident Reporting with Location
4. Emergency Help Requests
5. Volunteer / Community Assistance
6. Nearby Shelters & Emergency Services
7. Official Announcements & Updates
8. Map-based Disaster Tracking

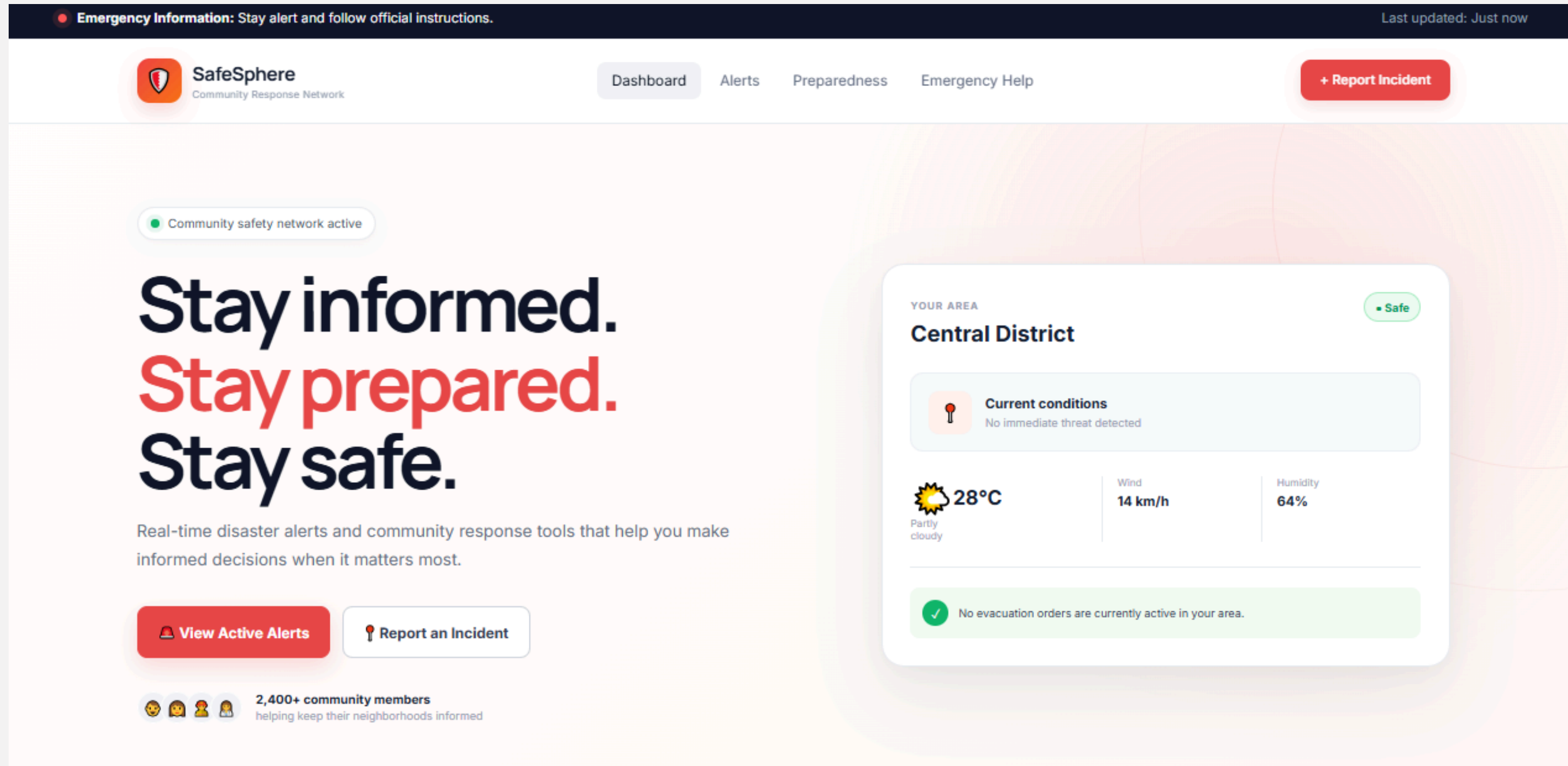


Frontend Design



- Home Page – introduces the platform and major emergency actions.
- Login / Registration – provides user authentication.
- Forms – used for incident reports and help requests.
- Dashboard – displays alerts, reports, assistance and updates.
- Navigation should make emergency actions easy to access.

Frontend Implementation



 SafeSphere
Community Response Network

DashboardAlertsPreparednessEmergency Help

+ Report Incident

 3
Active Alerts

 18
Incidents Reported

 94%
Areas Currently Safe

 2.4K+
Community Members

LIVE MONITORING

Active Disaster Alerts

Important safety information from your community and official monitoring sources.

 **Flood Warning** High

Riverside District

Heavy rainfall has caused rising water levels. Residents near low-lying areas are advised to move to safer locations.

 12 min ago Details →

 **Severe Weather** Medium

North Valley

Strong winds and thunderstorms are expected during the next few hours. Avoid unnecessary travel.

 28 min ago Details →

 **Earthquake Advisory** Low

Central Region

Minor tremors have been reported. No major damage has been confirmed. Stay alert for official updates.

 1 hr ago Details →

[View all alerts →](#)

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 01

Get Alerts

Receive timely information about disasters and hazards in your area.

 02

Report Incidents

Help your community by reporting flooding, fires, damage, or other hazards.

 03

Respond Together

Connect communities with emergency resources and response information.

BE PREPARED

Emergency Preparedness

A few minutes of preparation today can make a huge difference during an emergency.

 **Emergency Kit**

Learn what to keep ready for the first 72 hours.

[Learn more →](#)

 **Prepare Your Home**

Simple steps to make your home safer before a disaster.

[Learn more →](#)

 **First Aid Basics**

Essential first-aid information for emergency situations.

[Learn more →](#)

 **Emergency Contacts**

Keep important emergency numbers easily accessible.

[Learn more →](#)

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COMMUNITY RESPONSE

When an emergency happens, every second matters.

Quickly connect with emergency services, notify your community, or report hazards around you.

 Request Emergency Help

Report a Hazard



HOW IT WORKS

One platform. Stronger communities.

Simple tools designed to help people respond quickly and responsibly.

 SafeSphere
Community Response Network

DashboardAlertsPreparednessEmergency Help

+ Report Incident

COMMUNITY VOICE

Recent Community Reports

Information shared by people helping monitor their neighborhoods.

 **Road flooding reported** ✓ Verified

Riverside Road • 8 minutes ago

 **Power outage in residential area** Investigating

North Valley • 23 minutes ago

 **Fallen tree blocking road** ✓ Verified

West Avenue • 41 minutes ago

 **Keep your community informed.**

Share verified information and help others stay prepared.

[Report an Incident →](#)

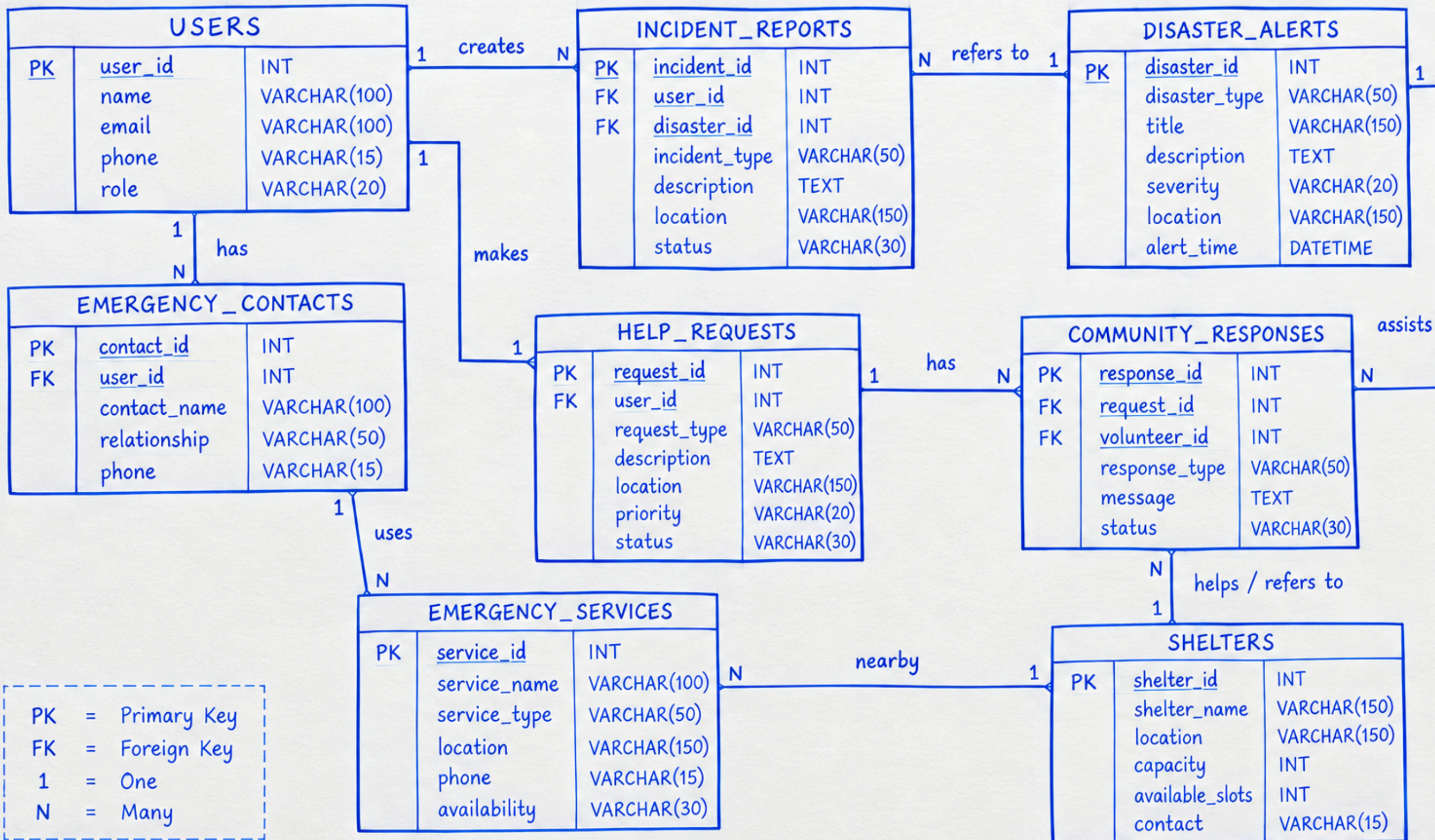
Database Design

- Users and emergency contacts
- Disaster alerts
- Incident reports
- Emergency/help requests
- Community or volunteer responses
- Shelter and emergency-service information



ER Diagram

ER Diagram - Disaster Alert & Community Response App



Database Tables / Sample Records

	contact_id	user_id	contact_name	relationship	phone
▶	1	1	Suresh Kumar	Father	9988776655
	2	2	Lakshmi Reddy	Mother	9988776656
	3	3	Meena Sharma	Mother	9988776657
	4	4	Raj Singh	Brother	9988776658
	5	5	Anil Rao	Father	9988776659
✱	NULL	NULL	NULL	NULL	NULL

	disaster_id	disaster_type	title	description	severity	location
▶	1	Flood	Heavy Flood Alert	Water levels are rising rapidly	High	Vijayawada
	2	Cyclone	Cyclone Warning	Strong winds expected in coastal areas	High	Visakhapatnam
	3	Fire	Industrial Fire	Fire reported near industrial area	Critical	Guntur
	4	Earthquake	Earthquake Alert	Earthquake reported in the region	Medium	Hyderabad
⌵	NULL	NULL	NULL	NULL	NULL	NULL

[illegible]

Current Progress

- Project title, abstract and key features defined.
- Frontend structure planned for home, authentication, forms and dashboard.
- Backend technology identified: Node.js.
- Database technologies identified: MySQL and MongoDB.
- REST API / OpenAPI identified for backend communication.
- Modules and architecture identified.
- Add screenshots/evidence for modules already implemented.

Challenges / Issues Faced

- Designing a simple interface for emergency situations.
- Managing location-based incident information.
- Keeping alerts and reports organized.
- Connecting frontend forms with backend services.
- Designing suitable database relationships.
- Ensuring authentication and emergency data are handled securely.

Future Work

- Implement and integrate all planned modules.
- Complete REST APIs and backend services.
- Add real-time alert delivery.
- Improve map-based disaster tracking.
- Integrate nearby shelters and emergency services.
- Strengthen authentication and authorization.
- Containerize and deploy using Docker.

Conclusion

- The proposed app provides a centralized platform for disaster communication.
- It combines alerts, incident reporting, help requests and community assistance.
- The planned architecture supports scalable and maintainable development.
- The project aims to improve reporting speed, coordination and community response.
- Next step: complete integration and demonstrate the working modules.